

FREDERICTON INTL, NB, Canada

WMO: 717000

Lat: 45.869N

Lon: 66.537W

Elev: 21

StdP: 101.08

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.1	-20.3	-29.1	0.3	-21.7	-26.6	0.3	-19.6	11.4	-2.4	10.3	-3.2	2.4	230	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	29.9	21.0	28.1	20.0	26.5	19.1	22.4	27.9	21.4	26.2	20.4	24.6	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	15.3	24.8	19.7	14.5	23.8	18.8	13.7	22.8	66.0	27.7	62.3	26.2	59.0	24.7	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.7	7.7	-27.2	33.3	2.9	1.3	-29.3	34.2	-31.0	35.0	-32.6	35.7	-34.7	36.7		
(5)			-27.5	24.2	2.8	0.9	-29.5	24.9	-31.1	25.4	-32.7	25.9	-34.7	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-8.6	-7.2	-1.8	4.9	11.4	16.6	20.0	19.1	14.7	8.2	2.1	-4.4		
	DBStd	10.84	6.60	6.09	5.20	3.84	3.65	3.55	2.70	2.85	3.92	4.06	4.91	5.67		
	HDD10.0	2390	578	483	368	159	24	1	0	0	5	84	242	446		
	HDD18.3	4541	836	716	626	404	216	75	14	26	121	316	488	704		
	CDD10.0	1039	0	0	1	5	69	197	308	281	147	28	3	0		
	CDD18.3	149	0	0	0	0	2	22	64	48	13	1	0	0		
	CDH23.3	1343	0	0	1	3	59	234	538	395	108	4	0	0		
	CDH26.7	333	0	0	0	1	13	63	132	97	28	0	0	0		
(14) Wind	WSAvg	3.4	3.5	3.6	4.0	3.9	3.7	3.3	2.9	2.7	3.0	3.3	3.4	3.5		
(15) Precipitation	PrecAvg	1151	105	87	96	86	96	85	88	91	95	97	111	116		
	PrecMax	1520	223	164	194	155	211	199	174	238	271	182	206	206		
	PrecMin	845	20	28	27	20	20	26	29	17	20	21	44	48		
	PrecStd	162	43	35	36	34	47	41	35	51	45	43	39	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	9.4	14.9	21.1	28.4	31.4	32.0	31.8	30.2	23.3	17.2	11.7		
		MCWB	10.8	7.6	8.6	11.2	17.9	21.2	22.4	22.1	21.4	16.6	15.4	10.9		
	2%	DB	6.9	5.7	9.6	16.9	24.7	28.4	29.7	29.3	26.3	19.6	14.7	8.4		
		MCWB	6.1	4.0	5.6	9.6	15.4	19.3	21.2	20.9	19.4	15.0	12.8	7.5		
	5%	DB	3.4	3.6	7.2	14.1	21.6	26.0	28.0	27.2	23.7	17.2	12.3	5.6		
		MCWB	2.2	2.0	4.3	8.8	13.8	17.9	20.3	19.9	18.1	14.0	10.3	4.3		
	10%	DB	0.8	1.7	5.2	11.8	19.1	23.7	26.5	25.4	21.6	15.3	9.6	3.2		
		MCWB	-0.1	0.3	2.5	7.3	12.4	16.6	19.5	19.0	17.1	12.3	7.9	2.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	7.9	10.2	14.7	19.2	22.6	23.9	23.3	22.4	18.5	16.0	11.0		
		MCDB	11.0	9.2	11.9	18.3	25.8	29.9	29.7	28.7	28.7	20.7	17.0	11.7		
	2%	WB	6.2	4.5	6.8	11.8	16.9	20.8	22.5	22.1	20.7	16.5	13.3	7.8		
		MCDB	6.8	5.3	8.5	14.8	21.9	26.1	27.8	27.6	24.2	18.4	14.4	8.3		
	5%	WB	2.3	2.1	4.6	9.8	15.1	19.4	21.5	21.1	19.4	14.7	10.6	4.6		
		MCDB	3.2	3.2	6.7	13.1	19.5	23.7	26.3	25.5	22.3	17.1	11.8	5.3		
	10%	WB	0.0	0.4	2.9	7.7	13.5	18.1	20.5	20.1	18.1	12.7	7.9	2.1		
		MCDB	0.7	1.5	4.9	11.2	17.9	22.0	24.8	23.8	20.6	14.8	9.3	3.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.3	9.5	10.0	11.6	11.5	11.3	11.6	11.6	10.2	8.0	8.2		
		MCDBR	13.0	10.6	11.9	14.9	17.1	15.6	14.0	14.2	13.9	13.3	11.2	10.2		
		MCWBR	12.8	9.3	8.8	9.3	9.3	8.2	6.8	7.2	8.0	9.3	9.8	9.7		
		MCDBR	13.1	10.0	10.8	12.6	13.9	12.8	12.0	11.5	11.6	10.2	10.2	10.2		
	5% WB	MCWBR	13.1	9.3	8.8	9.2	8.5	7.5	6.3	6.6	7.3	9.2	9.9	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.286	0.317	0.348	0.374	0.407	0.426	0.400	0.369	0.328	0.306	0.287		
	taud		2.453	2.410	2.435	2.420	2.390	2.302	2.269	2.345	2.405	2.525	2.538	2.475		
	Ebn at Noon		842	900	916	911	894	861	838	848	846	841	800	785		
	Edh at Noon		68	86	97	108	116	127	130	116	100	77	63	61		
(44) All-Sky Solar Radiation	RadAvg		1.50	2.37	3.51	4.35	4.99	5.39	5.49	4.86	3.93	2.43	1.43	1.11		
	RadStd		0.16	0.21	0.23	0.41	0.57	0.41	0.44	0.31	0.29	0.23	0.16	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+52	N/A

Nomenclature: See separate page